

NGI FORGE

SOFTWARE DISTRIBUTION PLATFORM FOR NGI PROJECTS

NixCon 2026, Kraków, Poland

Nix@NGI Team presented by Ivan Mincik (@imincik)

INTRO

NGI (NEXT GENERATION INTERNET) PROGRAM

Internet of Humans - trust, security, and inclusion and other values and norms that we enjoy in Europe



<https://www.ngi.eu/about/>

NGI ZERO

A **consortium** of not-for-profit organisations from across Europe



<https://www.ngi.eu/ngi-projects/ngi-zero/>

NLNET FOUNDATION

NGI Zero consortium leader

1000+ funded free software projects (1800+ cascading grants)



<https://nlnet.nl/>

NIXOS FOUNDATION

NGI Zero consortium member



<https://nixos.org/>

NGI TEAM @ NIXOS FOUNDATION



NixOS Foundation deliverables:

- 1. Package and maintain all **NGI-funded projects** with Nix
- 2. Organize **Summer of Nix**

<https://nixos.org/community/teams/ngi/>

OUR CHALLENGES

#1

**How to package 1000+ projects with varying complexity, maturity,
and ecosystem support?**

#2

How to sustainably maintain 1000+ projects?

THE IDEA



Attract upstream developers/contributors and distribute the packaging and maintenance effort

NGI FORGE



KEY DESIGN PRINCIPLES

- Make packaging of software intuitive
- Provide attractive additional benefits for contributors
- Target wide range (also non-Nix) of users
- **Gradually expose** users/contributors to the super-powers of Nix

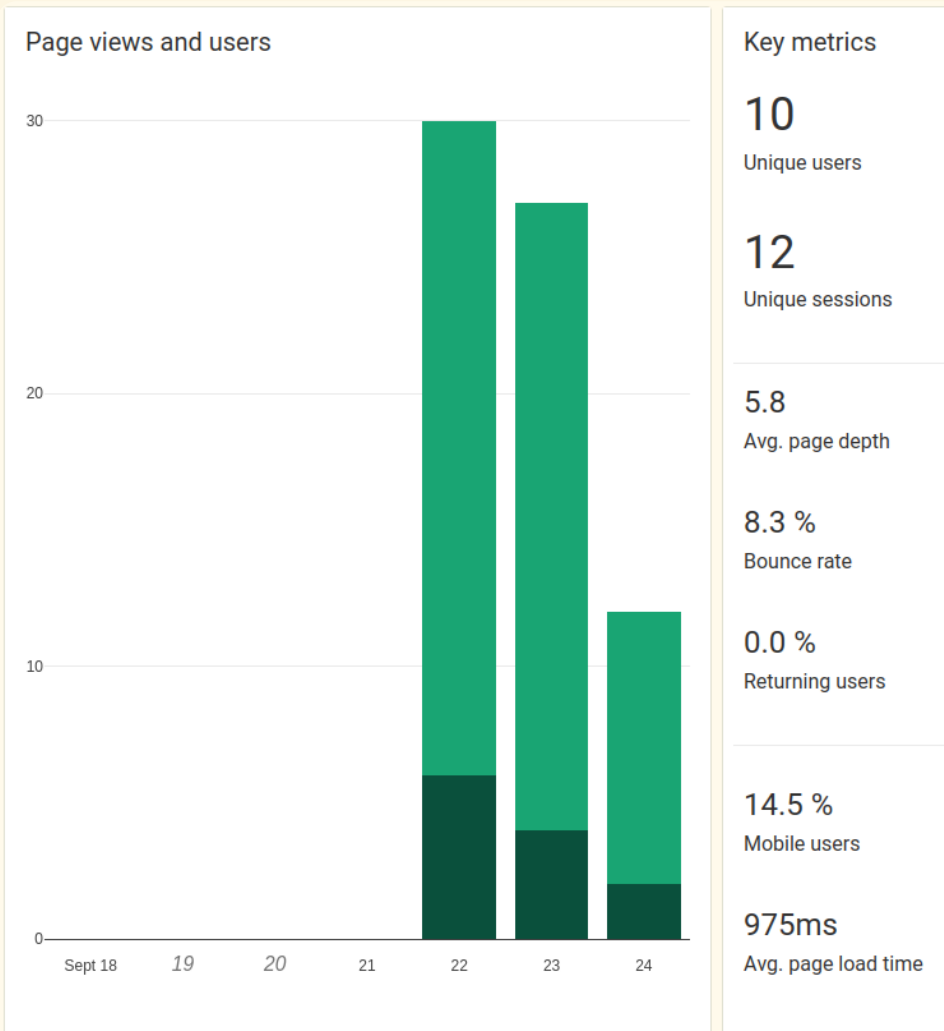
FEATURES

- Build software into **packages**
- Compose **applications** for multiple **runtimes**
 - Programs (CLIs, GUI): **program** and **shell**
 - Services: **container** and **NixOS**



BUILDING PACKAGES AND APPLICATIONS

OFFEN - FAIR WEB ANALYTICS



- 55 % Javascript + 40 % Go
- NGIO PET 2019 - 2019
- NGIO Discovery 2020 - 2022
- <https://github.com/offen/offen>

PACKAGES

`recipes/pkgs/offen/recipe.nix`

```
pkgs.offen = {  
  # package configuration  
  # ...  
};
```

METADATA AND SOURCE

```
1  pkgs.offen = {  
2    version = "1.4.2-unstable-2026-06-11";  
3    description = "Fair and privacy-focused web analytics.";  
4    homepage = "https://www.offen.dev";  
5    mainProgram = "offen";  
6  
7    source = {  
8      git = "github:offen/offen/ec99082a37ffb5855bd84debfe227d41c7b403c";  
9      hash = "sha256-EGlqD3611sG3YTVe74H49PB8Hj1NsKYhLANg5VAQ0wg=";  
10   };  
11   };
```

METADATA AND SOURCE

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1  pkgs.offen = {  
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9      hash = "sha256-EGlqD3611sG3YTVe74H49PB8Hj1NsKYhLANg5VAQ0wg=";  
10   };  
11   };
```

PACKAGE BUILDER

```
1  pkgs.offen = {  
2    # ...  
3  
4    build.goPackageBuilder = {  
5      enable = true;  
6      vendorHash = "sha256-AeQa5oa0EB/50aPCRq702vMEtEctwP+jU5C6zB+3XR0=";  
7      ldflags = [ "-s" "-w" ];  
8      modRoot = "server";  
9    };  
10 };
```

- npmPackageBuilder
- pythonAppBuilder
- pythonPackageBuilder
- rustPackageBuilder
- standardBuilder

TEST

```
1  pkgs.offen = {  
2    # ...  
3  
4    test = {  
5      script = "offen --help | grep <something>";  
6      runner = "bash"; # or nixos  
7    };  
8  };
```

BUILD AND TEST THE PACKAGE

Build package

```
nix build .#pkgs.offen
```

Run test

```
nix build .#pkgs.offen.test
```

INTERACTIVE PACKAGE DEBUGGING

```
build.debug = true; # recipes/pkgs/offen/recipe.nix
```

```
1  nix develop .#pkgs.offen
2
3  >>> Phase:  **unpackPhase** patchPhase updateAutotoolsGnuConfigScriptsPhase configurePhase buildPhase
4  >>> Command:  runPhase unpackPhase
5  >>> Press ENTER to run, CTRL-C to exit
```

INTERACTIVE PACKAGE DEBUGGING

```
build.debug = true; # recipes/pkgs/offen/recipe.nix
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```
1  nix develop .#pkgs.offen
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3  >>> Phase:  **unpackPhase** patchPhase updateAutotoolsGnuConfigScriptsPhase configurePhase buildPhase .
4  >>> Command:  runPhase unpackPhase
5  >>> Press ENTER to run, CTRL-C to exit
```

APPLICATIONS

`recipes/apps/offen/recipe.nix`

```
apps.offen = {  
  # application configuration  
  # ...  
};
```

METADATA

```
1  apps.offen = {  
2    displayName = "Offen";  
3    description = "Fair and privacy-focused web analytics.";  
4    longDescription = '  
5      Offen is a self-hosted web analytics server that gives operators insight  
6      into usage while allowing users to access, review, and delete their own  
7      data.  
8    ';  
9    usage = '  
10     ...  
11    '  
12  };
```

SERVICES

```
1  apps.offen = {
2    # ...
3
4    services = {
5      components.offen = {
6        process.command = pkgs.offen; # <-- Forge package
7        process.argv = [ "serve" ];
8        process.environment = {
9          OFFEN_SERVER_PORT = "3000";
10       };
11     };
12
13     runtimes = {
14       container.enable = true;
15       nixos.enable = true;
16     };
17   };
18 };
```

SERVICES

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1  apps.offen = {
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15       nixos.enable = true;
16     };
17   };
18 }
```

TEST

```
1 apps.offen = {  
2   # ...  
3  
4   test.services.script = '  
5     curl localhost:3000 | grep "Offen Fair Web Analytics"  
6   ';  
7 };
```

Run test

```
nix build .#apps.offen.test
```

RUN A SERVICE IN SANDBOX (BUBBLEWRAP)

```
1  nix run .#apps.offen.services.offen
2
3  [2026-09-22T07:50:45Z INFO  nimi::cli] Launching process manager...
4  [2026-09-22T07:50:45Z INFO  nimi::process_manager] Starting process manager...
5  [2026-09-22T07:50:45Z INFO  offen] Environment: OFFEN_DATABASE_CONNECTIONSTRING=/var/lib/offen/offen.d
6  [2026-09-22T07:50:45Z INFO  offen] Environment: OFFEN_DATABASE_DIALECT=sqlite3
7  [2026-09-22T07:50:45Z INFO  offen] Environment: OFFEN_SERVER_PORT=3000
8  [2026-09-22T07:50:45Z INFO  offen] Environment: PATH=/nix/store/9394ss2hhwhslkvf9bva1sh8bl2v1ifa-nimi-
9  [2026-09-22T07:50:45Z INFO  offen] Environment: PWD=/var/lib/offen
10 [2026-09-22T07:50:45Z INFO  offen] Environment: SHLVL=0
11 [2026-09-22T07:50:45Z INFO  offen] Environment: XDG_CONFIG_HOME=/tmp/nimi-config-44136fa355b3678a1146a
12 [2026-09-22T07:50:45Z INFO  offen] XDG_CONFIG_HOME=/tmp/nimi-config-44136fa355b3678a1146ad16f7e8649e94
13 [2026-09-22T07:50:45Z INFO  offen] Running: /nix/store/xnjs6nb1rj4qvf0b1a1ck96g0n7l6rz3-offen-1.4.2-ur
```

RUN A SERVICE IN SANDBOX (BUBBLEWRAP)

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13 [2026-09-22T07:50:45Z INFO  offen] Running: /nix/store/xnjs6nb1rj4qv0b1a1ck96g0n7l6rz3-offen-1.4.2-ur
```



WEB USER INTERFACE

NGI APPLICATIONS CATALOGUE

NGI Forge Search applications

Packages Options Docs

96 applications Sort: Random

Prev 1 / 8 Next

Application	Description	Tags
Alive2	Automatic verification of LLVM optimizations.	shell
Bottles	Easy-to-use wineprefix manager.	shell
cdxgen	Tool to create CycloneDX Software Bill-of-Materials (SBOM) for projects from source and container images.	program, shell, container, nixos
Kaidan	User-friendly and modern chat app, using XMPP.	program, shell
Offen	Fair and privacy-focused web analytics.	container, nixos
PyCM	Machine learning post-processing and analysis library to evaluate algorithm performance.	shell
Dino	Open source XMPP messaging application.	program
TeaType	Real-time co-editing of local text files.	program, shell

<https://ngi.nixos.org/>

OFFEN

 **NGI Forge**

Search applications

Packages Options Docs

Resources

Homepage

Documentation

Source Repository

Forge Recipe

NGI Grants

Review

offen

OffenOne

Configuration

Services

• offen

3000 → 3000

Runtimes

container nixos

Tried this app? We'd love to [know how it went.](#)

 **Offen**

Run Deploy

Offen is a self-hosted web analytics server that gives operators insight into usage while allowing users to access, review, and delete their own data.

Usage Instructions

First, launch the app [in a container](#) or [in a NixOS VM](#).

Initial Setup

1. Enter the running container

```
podman exec -it offen_offen_1 bash
```

Copy

or log in to the NixOS VM (the console logs in automatically as root).

2. Create an operator account

```
export OFFEN_DATABASE_CONNECTIONSTRING="/var/lib/offen/offen.db"
offen setup -name <account-name> -email <email> -password <password>
```

Copy

RUN OFFEN IN CONTAINER

NGI Forge

Resources

- Homepage
- Documentation
- Source Repository
- Forge Recipe

NGI Grants

- Review
 - offen
 - OffenOne

Configuration

Services

- offen
3000 → 3000

Runtimes

- container nixos

Offen

Container NixOS

Install Nix

Flakes Recommended Traditional Docs

Run application services in OCI containers

```
nix run github:ngi-nix/forge/66ba1bec#apps.offen.container
```

Copy

► Build container images manually

or log in to the NixOS VM (the console logs in automatically as root).

2. Create an operator account

```
export OFFEN_DATABASE_CONNECTIONSTRING="/var/lib/offen/offen.db"
offen setup -name <account-name> -email <email> -password <password>
```

Copy

RUN OFFEN IN CONTAINER

```
1 nix run github:ngi-nix/forge/f00365b4#apps.offen.container
2
3 Creating container image /home/imincik/.cache/ngi-forge/e70cdb81/offen-offen-j3g9by1lj3mk3p9yclhzpbx15y9y
4 Loaded image: localhost/offen-offen:j3g9by1lj3mk3p9yclhzpbx15y9y36rr
5
6 [offen] | [2026-09-03T09:36:59Z INFO nimi::cli] Launching process manager...
7 [offen] | [2026-09-03T09:36:59Z INFO nimi::process_manager] Starting process manager...
8 [offen] | [2026-09-03T09:36:59Z INFO offen] Running: /nix/store/hglbmavg7kjpvp36yy247dzms37xk53h-offen-1
```

 http://localhost:3000 

 Offen Fair Web Analytics

We only access usage data with your consent.

Offen Fair Web Analytics is a fair and open source analytics software. Help the website that linked you here by allowing access to your usage data.

Your data always remains yours. You can review and delete it at any time. Opt out again whenever you want.

I allow

I don't allow

How can I review and delete my usage data?

For data to be collected you need to opt in first. **You can do this in the consent banner.**

What data is collected?

Only page views, sessions, referrer and visited URLs are collected. From this, other metrics such as the bounce rate are calculated.

Why do I need to opt in or opt out?

Our banner serves two purposes: We ask you to access your usage data and would like to obtain your consent for its use.

Do you use cookies?

Yes, in particular cookies are used to store your decision about granting access to your usage data. If you opt in, we also use cookies to store a user and a session ID for you.

RUN OFFEN IN CONTAINER

```
1 nix run github:ngi-nix/forge/f00365b4#apps.offen.container
2
3 Creating container image /home/imincik/.cache/ngi-forge/e70cdb81/offen-offen-j3g9by1lj3mk3p9yclhzpbx15y9y
4 Loaded image: localhost/offen-offen:j3g9by1lj3mk3p9yclhzpbx15y9y36rr
5
6 [offen] | [2026-09-03T09:36:59Z INFO nimi::cli] Launching process manager...
7 [offen] | [2026-09-03T09:36:59Z INFO nimi::process_manager] Starting process manager...
8 [offen] | [2026-09-03T09:36:59Z INFO offen] Running: /nix/store/hglbmavg7kjpvp36yy247dzms37xk53h-offen-1
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ADDITIONAL BENEFITS

FORGE BUILT-IN PACKAGE DEVELOPMENT ENVIRONMENT

```
nix develop .#pkgs.offen.env
```

```
cp -r --no-preserve=mode,ownership  
/nix/store/agx01h4fykiwmy7p18w0a50rzpqhcg9n-source  
src
```

```
cd src/server
```

```
go build -o offen ./cmd/offen
```

```
./offen demo
```

WIP: BENEFITS FOR UPSTREAM DEVELOPERS

Integrate Forge package

```
nix flake init --template github:ngi-nix/forge#developer
```

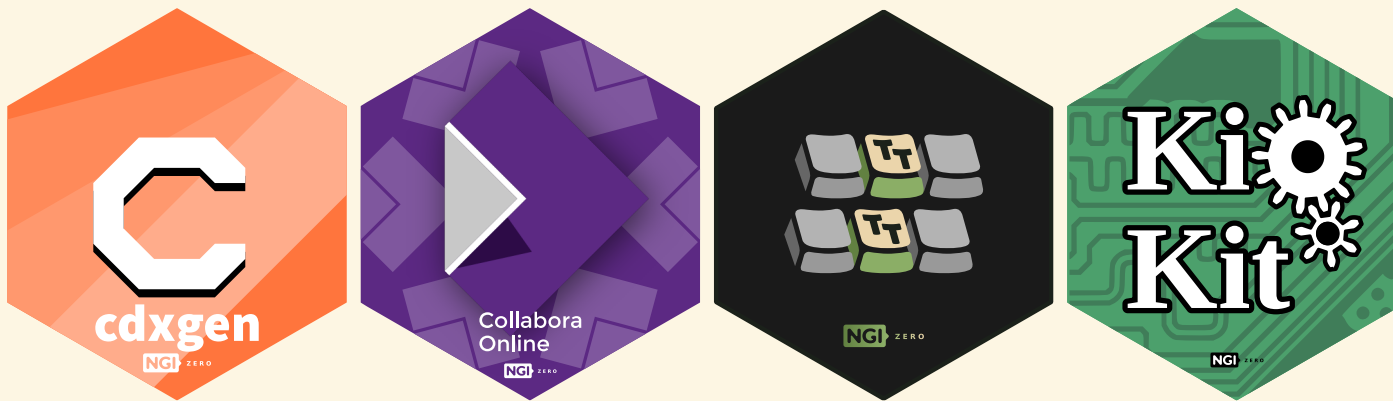
Launch development environment

```
nix develop
```

Run software from upstream repo

```
nix run github:<owner>/<repo>#package
```

NIXPKGS



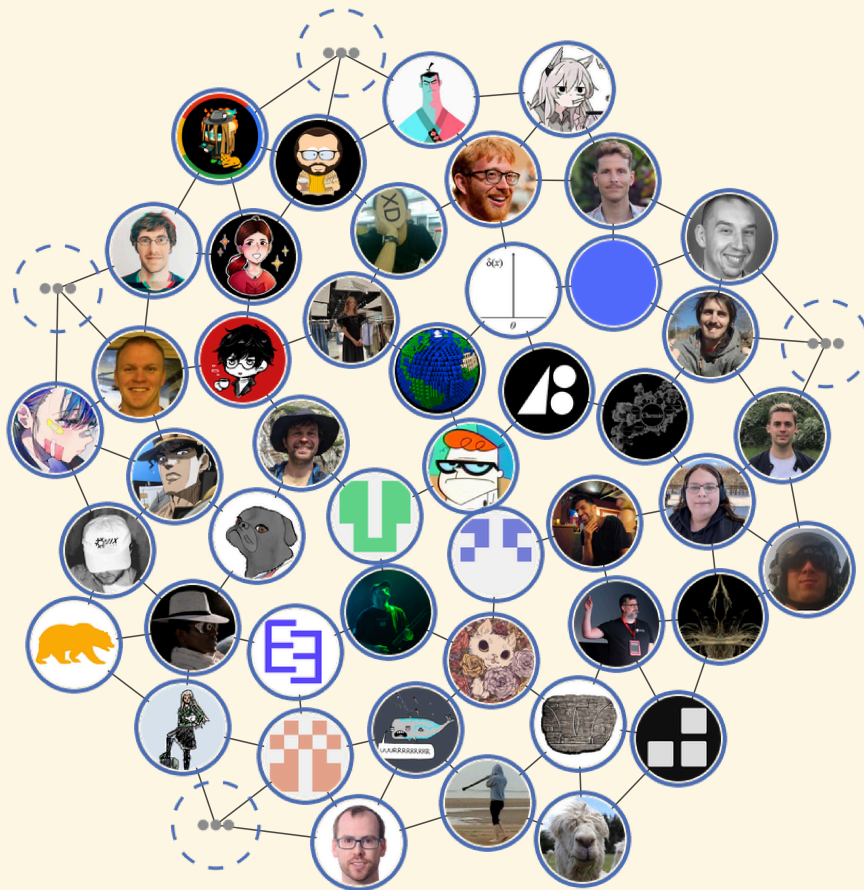
> 300 packages maintained by NGI Team

HOW TO CONTRIBUTE

- Matrix room
- NGI projects board

<https://nixos.org/community/teams/ngi/>

THANK YOU



<https://ngi.nixos.org/>